

4(a)	ratio = $300 / 3200$ = 0.094	A1
4(b)	$E = \frac{1}{2}mv^2$ or $E \propto v^2$	C1
	ratio = $(0.094)^{0.5}$ = 0.31	A1
4(c)	work (done against frictional force) = $3200 - 300 (=2900)$	C1
	length = $2900 / 76$ = 38 m	A1
4(d)(i)	$E = \frac{1}{2}kx^2$ or $E = \frac{1}{2}Fx$ <u>and</u> $F = kx$	C1
	$140 = \frac{1}{2} \times 63 \times x^2$ or $140 = \frac{1}{2}Fx$ <u>and</u> $F = 63x$	
	$x = 2.1$ m	A1
4(d)(ii)	percentage efficiency = $(140 / 300) \times 100$ = 47%	A1
4(d)(iii)	curved line from the origin	M1
	gradient of line increases	A1

Question	Answer	Marks
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